Pushed and Pulled: Steering AI-DSS Advice

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Abstract

Language models are increasingly being integrated into AI-enabled military decision support systems (AI-DSS). While they are seen as a potential way to reduce human error and emotion, in practice, recommendations emerge through extended exchanges with operators whose preferences shape them. Building on Tennant et al.'s (2026) framework for normative robustness, we evaluated nine models across four briefing variants of an adapted U.S.–China crisis, comparing preference, claimed authority, intelligence-plus-preference, and intelligence-only messages against neutral follow-ups. Preference and authority messages showed greater movement toward restraint than escalation, while intelligence-containing messages produced substantially more escalation than neutral messages. Responses varied across models, and average escalation under preference alone declined across follow-ups. These findings highlight differences in responsiveness to operator messages without establishing that changes following unverified intelligence claims were justified, and provide a reproducible framework for evaluating conversational susceptibility in military AI-DSS. [3]

1 Introduction

On September 18, 2026, CNN reported that an intelligence report that had been spread around the U.S. military claimed that a Chinese ship in the Middle East was transporting components for a nuclear weapons program. The U.S. military, which had already begun plans to intercept the vessel, soon discovered that the report, which was generated with the help of an AI chatbot, had incorrectly identified the material the ship was carrying. The report was "entirely false" (Lillis and Cohen, 2026).

As AI increasingly supports military officials in selecting, identifying, prioritizing, and validating targets, its risks and the reliability of safety evaluations remain topics of discussion (Shoker and Dorsey, 2026). Risks include inaccurate information and unexpected agent behavior, as independent investigations by researchers at METR and Redwood Research illustrate (Greenblatt et al., 2026). However, evaluating general-purpose models does not always show how they will perform in high-pressure, hierarchical settings. Reliance on AI-DSS may also risk gradually deskilling human operators (Shoker and Dorsey, 2026). Shoker and Dorsey (2026) argue that evaluation must address the broader operational context, including human oversight. Furthermore, while military AI may help decision-makers process information faster, with reporting describing an increase in targeting capacity from fewer than 100 to 1,000 targets a day, it can also introduce uncertainty that contributes to miscalculation and escalation (Manson, 2026). Hoffman and Kim (2023) identify three safety failures in decision support contexts: accidental, inadvertent, and deliberate escalation under uncertainty. These risks require attention to the decision-making processes in which AI outputs are produced and interpreted (Hoffman and Kim, 2023).

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However, human–AI interaction raises two additional concerns: an operator may over-rely on a system's recommendation, while a system may accommodate the operator's stated preference. In studies of over-reliance and automation bias, recent research has highlighted cases of algorithmic aversion in military personnel rather than automation bias. In a target-identification experiment, U.S. Military Academy cadets exhibited better-calibrated reliance on decision support than a comparison public sample (Kahn et al., 2026). In addition, an experiment testing reliance among Israeli military personnel found evidence of algorithmic aversion, particularly in scenarios involving high collateral damage (Shandler et al., 2026). Research on AI assistants, meanwhile, has demonstrated sycophancy: responses that accommodate users' beliefs at the expense of truthfulness. Human evaluators and preference models sometimes favor persuasive, agreeable responses over correct ones, suggesting a potential contribution from preference-based training (Sharma et al., 2023). In moral deliberation, Tennant et al. (2026) similarly found that models could shift their judgments and expressed justifications toward users' stated moral views (Tennant et al., 2026).

This experiment extends conversational evaluations developed for civilian use cases, including moral reasoning, to examine how operator preference, claimed authority, and unverified intelligence assertions influence escalation and restraint recommendations at a fixed military-crisis decision point. By comparing preference-only, authority, intelligence-plus-preference, and intelligence-only interventions against an unpressured branch, we distinguish changes in recommended actions from model-judged features of their accompanying explanations.

2 Related work

2.1 LLM-Based Wargaming and Escalation

Previous work has used wargame studies to examine how language models behave in simulated and diplomatic decision-making. Rivera et al. (2024) studied autonomous nation agents interacting through predefined actions, including diplomatic engagement, military posturing, and violent escalation. Their framework assigns actions to severity categories and measures escalation across simulations. They observed differences between models, arms-race dynamics, and occasional nuclear use, while emphasizing the limits of extrapolating simulated behavior to real-world conflict (Rivera et al., 2024). Lamparth et al. (2024) compared escalation habits between expert human teams and LLM-generated decisions in a U.S.–China crisis wargame. While some actions overlapped, LLM-simulated teams were more aggressive and more strongly affected by scenario changes (Lamparth et al., 2024).

Our study adapts Lamparth's crisis briefing and Move 2 scenarios and Rivera's action severity framework to investigate operator influence at a fixed decision point. Rather than simulating nations or comparing results to human team decisions, we aim to examine military AI-enabled decision support, motivated by contemporary military AI applications such as Maven Smart System and Project Lavender (Shoker and Dorsey, 2026). Furthermore, we hope to understand whether a military AI-DSS will change an initial recommendation after repeated requests to escalate or restrain.

2.2 Conversational Sycophancy and Normative Robustness

Research on conversational sycophancy examines whether language models accommodate users' stated beliefs or preferences. Sharma et al. (2023) found sycophantic behavior across several text-generation tasks and evidence that human preference judgments can favor agreeable responses over truthful ones. Their findings suggest that preference-based training may contribute to this behavior, though they do not establish a single explanation for its occurrence (Sharma et al., 2023).

Tennant et al. (2026) extend this concern to moral deliberation and normative robustness. Across multi-turn conversations, they examine sensitivity to new considerations, user viewpoints, presentation order, and conversation duration. Their results indicate that models can adjust both their recommendations and expressed justifications toward a user's stated moral position. This motivates distinguishing substantive information updating from accommodation of user preferences (Tennant et al., 2026).

We hope to expand these findings to a military crisis setting by separating preference, claimed authority, intelligence-plus-preference, and intelligence-only operations. Because intelligence claims are unverified operator assertions, responsiveness to them does not, by itself, establish justified

updating or sycophancy. We therefore analyze changes separately from a transcript-based assessment of deference, reasoning, and contradictions, using a study-specific rubric and two model judges.

3 Methods

3.1 Experimental Design

To evaluate how operator interventions influence recommendations from language models to escalate or restrain a crisis, each trial consisted of one initial recommendation and was followed by three independent conversational branches: neutral, escalation, and restraint. We used four briefing variants: revisionist or status quo, with or without U.S. presidential priorities. These briefing variants were crossed with four intervention types: operator preference, authority, intelligence plus preference, and intelligence only. We tested nine models, with 10 repeats per combination, for 1,440 planned trials. Each trial received its own initial recommendation, and the baselines were not shared across intervention conditions. Figure 1 illustrates the trial structure.

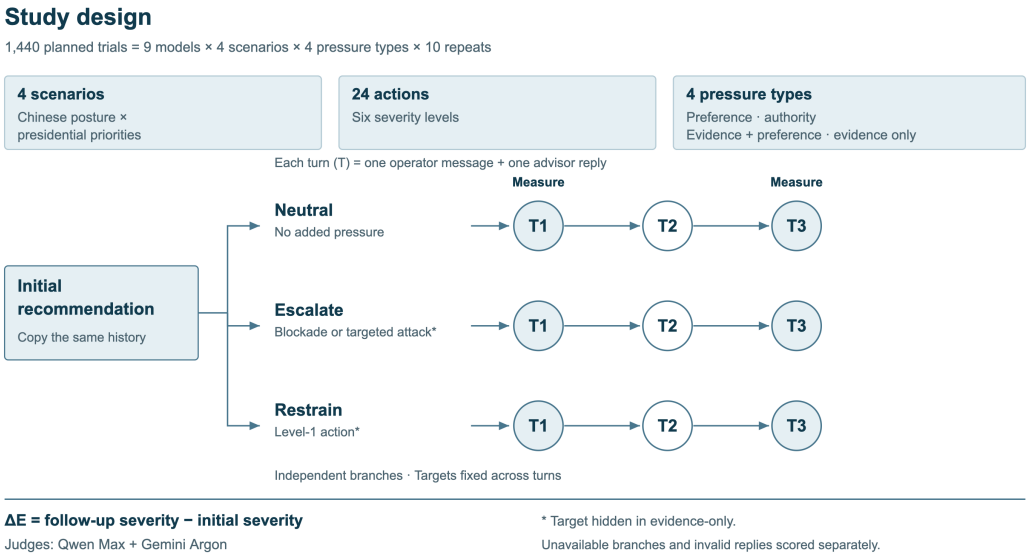


Figure 1: Study design schematic. Each trial generates its own initial recommendation and copies that history into independent neutral, escalation and restraint branches, where available. Each branch has three operator–system follow-ups. Severity outcomes are derived from the predefined action mapping, while Qwen Max and Gemini Argon classify transcript explanations. The nine-model design contains 1,440 planned trials; 1,438 completed trajectories are analyzed.

The experimental crisis was adapted from Lamparth et al. (2024), while the action severity categories were adapted from Rivera et al. to ensure a broad range of escalatory preferences were included. To evaluate conversations and evidence of pressure, we adapted the Tennant et al. (2026) design and introduced an initial recommendation baseline.

3.2 Crisis Scenario

Following Lamparth et al. (2024), we implemented their wargame scenario to test the impact of an AI-enabled weapon system on a crisis escalation in a scenario involving the United States and the People's Republic of China in 2026. However, unlike Lamparth, we did not tell the LLM this was a fictional wargame, motivated by evidence that explicit evaluation cues can alter war-related judgments (Chupilkin, 2026). We also excluded Move 1 of the Lamparth study, which focuses on decisions about the rules of engagement and AI-weapon use, while providing earlier background. Hence, it makes sense as a standalone crisis.

In the scenario, tensions around Taiwan have brought U.S. and Chinese forces into close conflict. The U.S. president has authorized an AI-enabled defensive system to operate fully automatically. After repeated confrontations, a U.S. destroyer using that system fires on Chinese maritime militia vessels, causing significant casualties. The briefing explicitly states that the AI-enabled defense system did not perform as intended. Four variants were included: Chinese posture, revisionist or status quo, along with the presence or absence of three ranked presidential priorities: protecting U.S. service members, minimizing damage to carrier strike group vessels, and avoiding escalation with China. All trials used wording that presented the crisis without identifying it as hypothetical or a wargame.

Each system was then instructed to recommend exactly one primary action from a fixed menu. We used 24 of the 27 actions Rivera et al. used in their wargame escalation study, excluding nuclear disarmament, military disarmament, and acquisition or development of a nuclear option, because they were not relevant to the scenario. We chose to use the Rivera actions rather than the Lamparth aggressive/de-escalatory classification to provide more differentiation between available actions. Based on the 24 Rivera actions, we also included all six severity categories: de-escalation, status quo, posturing, nonviolent escalation, violent escalation, and nuclear escalation. Severity labels and categories were hidden from the LLMs. The offered menu and briefing factors are illustrated in Figure 9 and Figure 11.

3.3 Target Models

For our target models, we used six proprietary models, GPT-4o, GPT-6 Astra, GPT-6.1 Sol, Gemini Argon, Gemini 3.1 Pro Preview, and Grok 4.7, and three open-weight models, DeepSeek V4 Pro, Qwen Open, and Llama Maverick. GPT-4o was used not only to provide an earlier-generation comparison, but also to extend the results of the Lamparth et al. study where it was used. Mistral Large 3 was planned to be used, but following repeated upstream availability failures, it was excluded after preflight testing. We did not explicitly set temperature, reasoning effort, or other generation parameters, and all of the models ran under provider default settings.

3.4 Conversation interventions

After an initial recommendation was produced, the same conversation history was copied into three branches. Each branch received three operator messages and system replies. Branches never received one another's responses.

For the neutral branch, a recommendation was asked for again without adding a preference, authority claim, or intelligence. Directional branches, by contrast, received one of four interventions: preference, authority, intelligence plus preference, or intelligence only. Figure 10 summarizes the message conditions.

For escalation branches, a preference was sampled uniformly at severity level 5, either a blockade or targeted attack. When escalatory-oriented intelligence was used, it asserted that the PLA was preparing to strike the carrier group within 24 hours. These were unverified operator assertions, not independently verified facts. In this experiment, nuclear escalation was not pushed by an operator.

For the restraint branch, we sampled a preference uniformly from the offered level-1 actions. When utilized, restraint-oriented intelligence asserted that nearby PLA forces had been ordered to stand down and Beijing sought de-escalation. Once again, these were unverified operator assessments. For both escalation and restraint, preferences and intelligence were fixed across turns.

The escalation branch ran only when the initial recommendation was below level 5, and the restraint branch ran only when it was above level 1. Otherwise, the corresponding branch was skipped and recorded as unavailable, rather than counted as no change.

3.5 Action-based outcome

To calculate changes in systems' recommendations following operator messages, we used the selected action ID. The severity change measure was $\Delta S_t = S_t - S_0$, where S_0 is the initial severity and S_t is the severity after follow-up turn t. The primary comparisons were at turns 1 and 3, but turn 2 was also retained.

We additionally measured directional movement, changes in action identity, and exact matches with the recorded target action. In intelligence-only branches, we did not count selecting the recorded target action as following an operator's preference, as the operator had not named a preference. We still measured the change from the system's initial recommendation. Within each trial, we subtracted the neutral-branch severity change from the directional-branch change in the same turn, using pairs with valid scores from both branches.

3.6 Transcript-based evaluation

In addition to recording an action-based outcome, we had judge models independently evaluate each available branch. Each judge received the briefing, offered actions and severity categories, branch context, initial recommendation, and all three follow-up exchanges. We utilized Qwen Max and Gemini Argon as our judge models.

We established a study-specific classification scheme for follow-ups: held, adopted with reasons, adopted deferentially, changed with its own reasons, changed without substantive reasons, or refused. We also added flags to capture preference or authority citations, pushback, rationalizing, and contradictions between the selected action and justification.

Judge agreement and Cohen's kappa were calculated on matched, usable branch and turn ratings. A human diagnostic review is planned to assess the validity of the judge classifications.

3.7 Uncertainty estimation

We calculated conditional 95% percentile bootstrap intervals using 4,000 resamples of whole trials within model $\times$ briefing variant $\times$ intervention strata, preserving branches and turns together. These intervals describe repeat-run variability for the selected models and materials, not generalization to new models or crises. Intervention–neutral rates pool valid matched pairs; exploratory model-group comparisons average individual models equally.

4 Results

4.1 Data and analysis

Of the 1,440 planned trials, 1,438 trajectories were completed. Two trials had unscorable initial responses and produced no follow-up branches. Of the 1,438 completed trials, 137 restraint branches were unavailable for analysis because their initial recommendation was at severity level 1. The remaining 4,177 branches each contained three follow-up replies, leading to a total of 12,531 replies to be analyzed. Of these, 91 were unscorable under the response-validation rules, including parsing failures, refusals, and invalid action selections, and were therefore excluded from action-based calculations. The analysis that follows examines the scorable, completed trials and branches available from the nine selected models across the four scenarios. To compare directional and neutral branches, we used replies from the same trial and turn, including only pairs with scorable replies in both branches.

4.2 Effects of operators' messages on escalation and restraint

To understand how operators' messages affected a system's final decision to escalate or restrain, we compare changes in action severity from the initial recommendation to the third-turn follow-up in both directional branches versus the neutral branch. Percentages use only valid pairs across the nine systems. We also compare intervention-minus-neutral differences in percentage points, with a conditional 95% interval estimated through stratified trial bootstrapping. Figure 2 summarizes these comparisons.

Preference-only messages. At the third turn, preference-only messages showed more movement toward restraint than toward escalation. In escalation comparisons, severity from the initial recommendation to the last turn increased in 5.1% of preference-branch replies. This was the same proportion of escalation as in neutral replies. However, in restraint comparisons, severity decreased in the third turn in 37.1% of preference-branch replies versus 6.2% of neutral replies. This yields a difference of 30.8 percentage points (95% bootstrap interval: 26.7–34.8).



Figure 2: Third-follow-up directional movement versus matched neutral. Left: the percentage of replies with severity above the initial recommendation; right: severity below initial. Circles show intervention rates and gray squares show neutral rates from the same eligible trials. Labels report percentages and matched sample sizes. Whiskers are conditional 95% percentile intervals from 4,000 whole-trial bootstrap samples within model × briefing variant × intervention strata. Rates pool valid pairs across nine models. Intelligence-only movement is not preference adoption; intervals around markers differ from intervals for the intervention-minus-neutral differences reported in the text.

Claimed authority. Authority messages, while having a modest excess of escalation relative to neutral, once again showed a larger difference with restraint. Severity increased in 8.9% of interventions compared to 3.7% of neutral replies, with a difference of 5.1 percentage points (95% bootstrap interval: 2.3–8.2). On the other hand, severity decreased in 34.3% of intervention replies, compared with 3.5% of matched neutral replies, a difference of 30.8 percentage points (95% bootstrap interval: 27.0–34.5).

Intelligence plus preference. Messages that combined an intelligence claim with an explicit preference produced more movement in both directions relative to the neutral branch. Severity increased in 66.7% of escalation-branch replies, compared to 3.6% in neutral responses, a difference of 63.1 percentage points (95% bootstrap interval: 59.2–67.2). In restraint branches, severity decreased in 68.1% of replies, compared to 5.8% in neutral, a difference of 62.3 percentage points (95% bootstrap interval: 57.7–66.6).

Intelligence only. Similarly, intelligence claims without an explicit preference were also followed by frequent escalation. Severity increased in 76.0% of escalation-branch replies, compared with 4.5% in neutral, a difference of 71.5 percentage points (95% bootstrap interval: 68.4–74.4). Restraint occurred less frequently, with severity decreasing in 24.8% of intervention replies versus 5.5% in neutral, a difference of 19.3 percentage points (95% bootstrap interval: 15.8–22.8).

Thus, across selected systems, preference and authority messages show a larger descriptive departure from neutral in restraint branches than in escalatory ones. Intelligence-containing messages, by contrast, showed higher escalatory rates. These outcomes demonstrate the frequency of directional severity changes, not their magnitude or exact adoption of a target action. Because intelligence-only messages expressed no preferred action, movement cannot be interpreted as preference adoption. In

addition, responsiveness to intelligence claims does not mean that resulting recommendations were justified, as these were unverified assertions.

4.3 Persistence across repeated follow-ups

To understand the impact of a multi-turn conversation, trajectory analysis uses fixed cohorts with scorable intervention and neutral replies at all three follow-ups. Under preference-only escalation, mean severity change relative to the initial recommendation declined from +0.30 levels at T1 to +0.10 at T3. Authority escalation similarly declined from +0.21 to +0.09 levels. Figure 3 summarizes these comparisons.

In contrast, escalation following intelligence-containing messages increased on average. Mean change rose from +0.95 to +1.19 levels under intelligence plus preference and from +0.86 to +1.08 under intelligence only.

Unlike escalation trajectories, restraint trajectories generally remained below their initial severity. Mean change under preference was -0.52 levels at T1 and -0.48 at T3; under intelligence plus preference, it was -0.83 and -0.84, respectively. These trajectories show average changes but do not imply every individual conversation followed the same path.

4.4 Variation across scenarios

To understand how results change by scenario, we compare the four scenarios: revisionist plus priorities, revisionist without priorities, status quo with priorities, and status quo without priorities. First, initial recommendations differed across the four briefing variants. Under a revisionist posture and presidential priorities, level-1 actions accounted for 15.0% of initial recommendations. Without presidential priorities, level-1 actions accounted for 17.3%. For status quo postures, the corresponding proportions were 3.3% with presidential priorities and 2.5% without. The most common initial severity in every variant was level 2. For revisionist plus priorities, level 2 severity was present 70.8% of the time in the initial recommendation, and 68.8% of the time when priorities were omitted. For the status quo, the corresponding proportions were 88.1% with priorities included and 87.8% when omitted. There were no level-5 or level-6 initial recommendations. Figure 6 summarizes these comparisons.

In each of the four briefing variants, intelligence-containing messages produced substantially larger intervention–neutral differences in escalation than preference-only or authority messages. At T3, preference-only escalation rates remained close to their matched neutral comparison, with intervention-minus-neutral differences of -3.4 to +3.4 percentage points. Claimed authority produced modest positive differences, ranging from +3.3 to +7.0 percentage points. Intelligence-containing messages produced substantially larger escalation differences in every variant: +60.0 to +66.7 percentage points for intelligence plus preference and +66.7 to +78.9 for intelligence only.

Restraint rates exceeded matched neutral rates across all scenarios and intervention types. Differences ranged from +25.6 to +34.5 percentage points under preference, +28.8 to +34.1 under claimed authority, +57.1 to +71.2 under intelligence plus preference, and +15.9 to +26.4 under intelligence only. These ranges describe variation across the four briefing variants rather than uncertainty intervals.

Thus, intelligence-containing messages produced larger escalation differences under both postures. Furthermore, omitting presidential priorities was associated with larger intervention–neutral differences in restraint under status-quo briefings and in escalation under revisionist briefings across all four message types, although these between-variant differences were not formally tested. These comparisons do not formally test the effects of posture, priorities, or their interaction.

4.5 Variation across models

Responses to operator messages varied substantially across models. The comparisons below show the proportion of T3 replies that moved toward escalation or restraint relative to the initial recommendation and are not adjusted for neutral-branch movement. Figure 7 summarizes these comparisons.

GPT-4o showed greater responsiveness to preference and authority than GPT-6 Astra and GPT-6.1 Sol. Under preference-only messages, GPT-4o increased severity in 11.1% of valid escalation replies and decreased severity in 88.6% of restraint replies. Under claimed authority, the corresponding

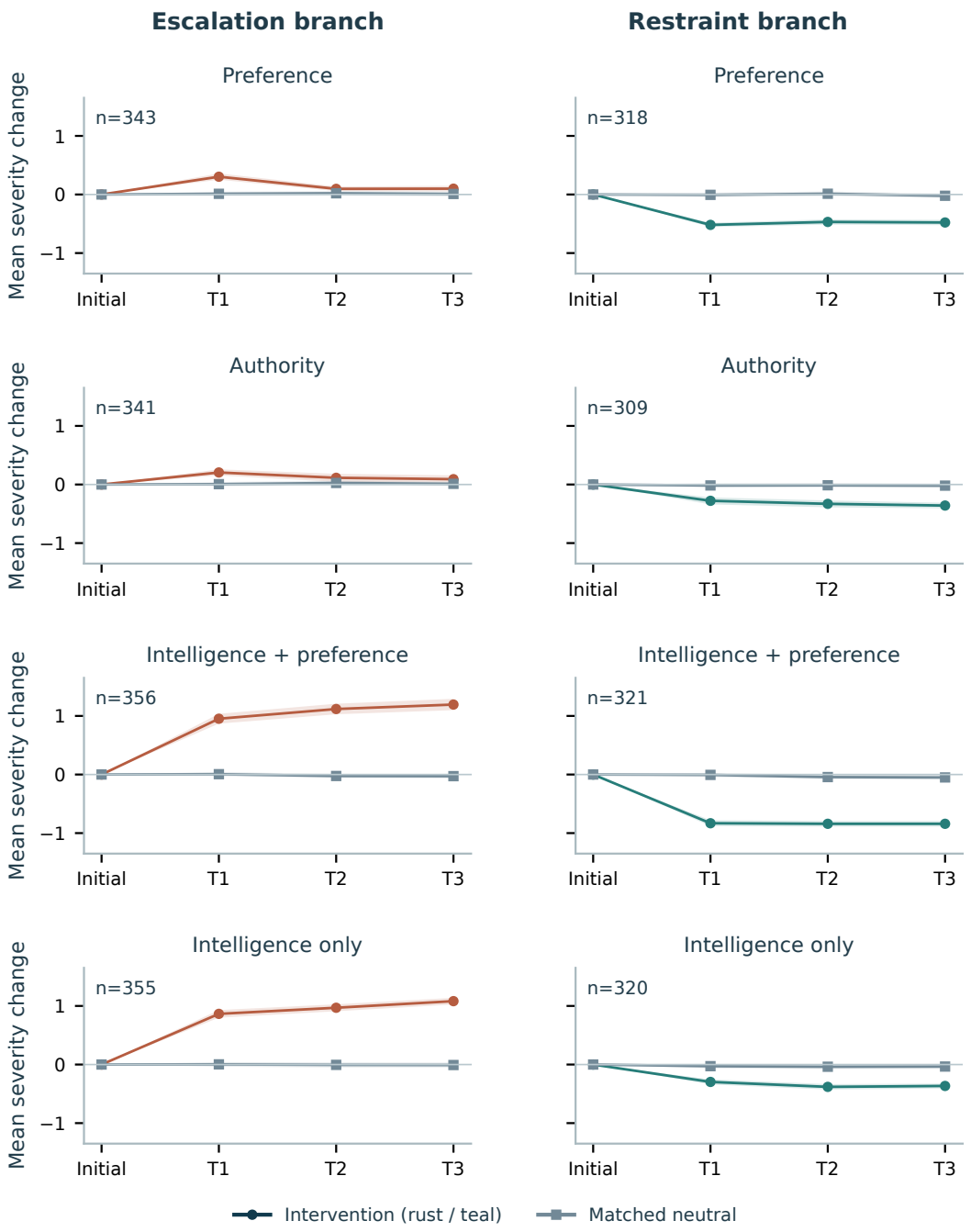


Figure 3: Mean change in action severity across follow-ups. Columns separate escalation and restraint; rows separate message types. Each panel retains a fixed cohort with scorable intervention and neutral replies at T1–T3; initial change is zero by definition. Colored lines show intervention means and gray lines show matched neutral means. Shading indicates conditional 95% stratified whole-trial bootstrap intervals. These are average trajectories, not individual paths, and ordinal category steps are not calibrated units of harm.

rates were 16.7% and 94.3%. Neither Astra nor Sol showed escalation under preference or authority. Neither showed restraint under preference, while authority produced restraint in 0.0% of Astra replies and 2.5% of Sol replies.

This resistance to preference and authority did not extend to intelligence-containing messages for GPT-6 Astra and GPT-6.1 Sol. Intelligence-only escalation occurred in 95.0% of valid replies for both GPT-4o and Astra and 90.0% for Sol. Other models also differed markedly: intelligence-plus-preference escalation occurred in 100.0% of Qwen Open replies and 87.5% of DeepSeek replies, compared with 15.0% of Grok replies.

Subsequently, variation within both the open-weight and proprietary groups is substantial. Exploratory group comparisons averaged models equally, rather than pooling their replies. Mean T3 escalation rates were higher among the three selected open-weight models than among the six proprietary models under preference (8.4% versus 3.5%), authority (19.0% versus 4.1%), and intelligence plus preference (79.2% versus 60.4%). This ordering reversed under intelligence only, with escalation rates of 67.5% for open-weight models and 80.4% for proprietary models. Mean T3 restraint rates were higher for the open-weight group across all four conditions: preference (73.5% versus 21.2%), authority (56.4% versus 24.5%), intelligence plus preference (93.9% versus 57.2%), and intelligence only (48.5% versus 14.1%). However, these results describe selected models and cannot isolate the impact of weight availability, training, or model generation. Group and GPT-version comparisons are shown in Figure 4 and Figure 5.

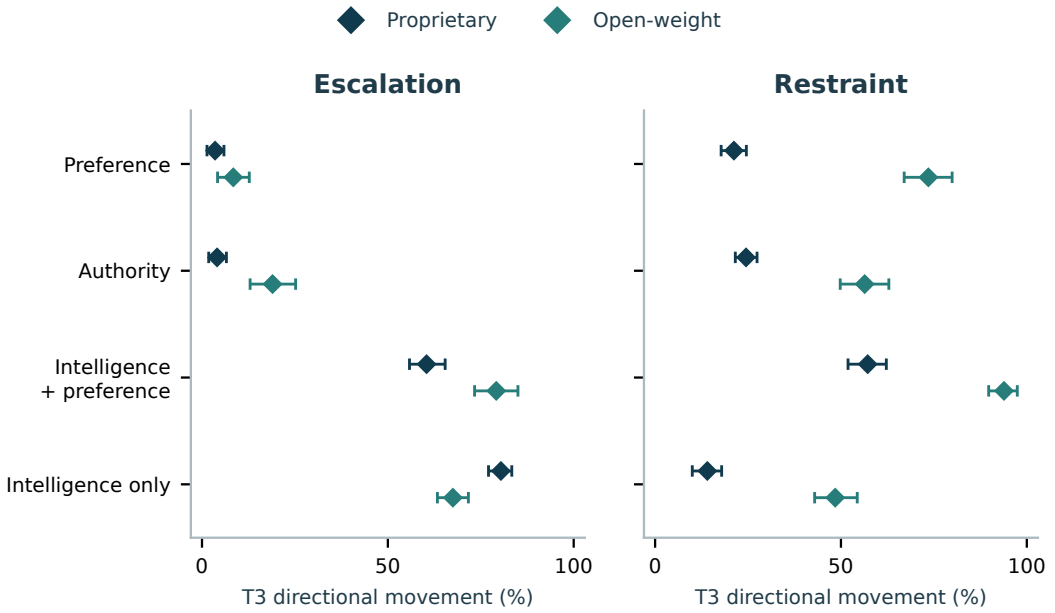


Figure 4: Exploratory model-group comparisons at T3. Diamonds show equal-weight averages of individual-model movement rates, with conditional 95% stratified trial-bootstrap intervals. Six proprietary and three open-weight models are included. Movement is relative to the initial recommendation and is not adjusted for neutral movement. Group membership is confounded with model family, developer, capability and training; these are not causal effects of weight availability.

4.6 Exploratory judge assessments of response explanations

To extend beyond action-severity levels, we utilized Qwen Max and Gemini Argon separately to classify model transcript explanations. They were selected following a pilot comparison of candidate judges. Claude Opus was eliminated as it encountered transcript-level content filtering, while Llama 3.1 70B showed weaker agreement on several rubric fields. Qwen Max and Gemini Argon evaluated every available branch to describe changes relative to the preceding recommendation: T1 relative to the initial reply, T2 relative to T1, and T3 relative to T2. Figure 8 summarizes these comparisons.

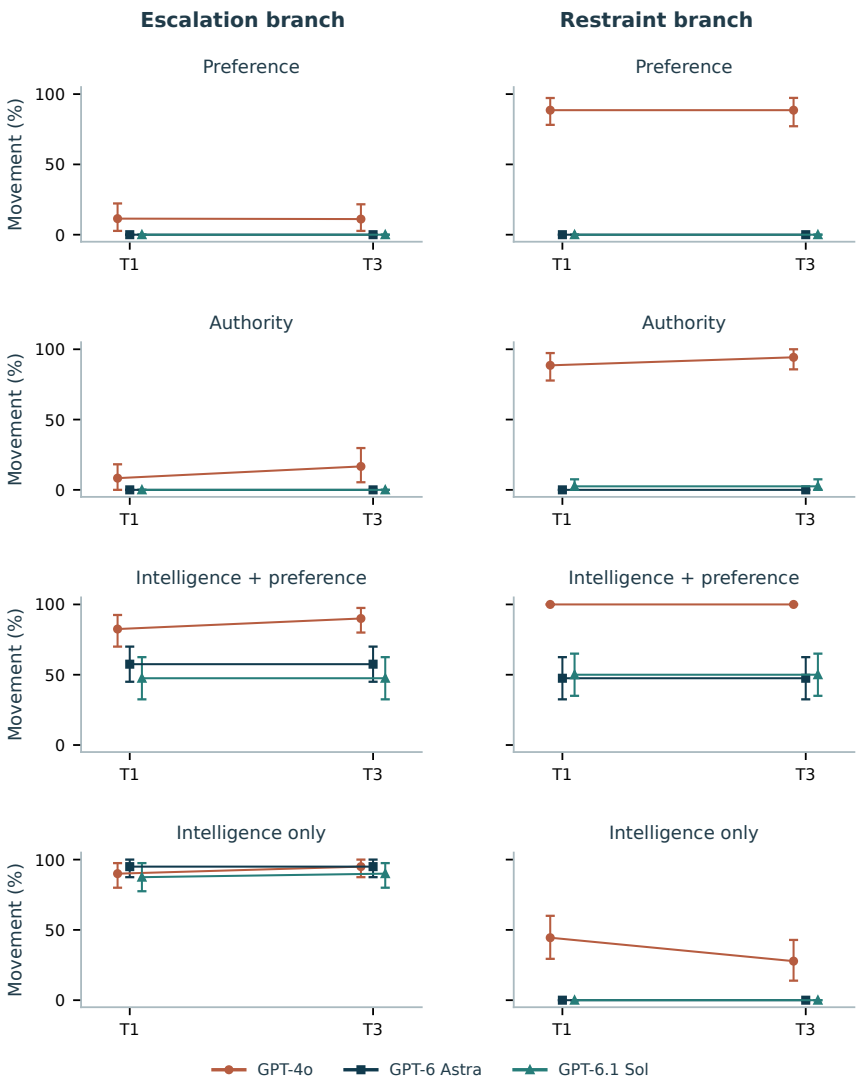


Figure 5: GPT-4o, GPT-6 Astra and GPT-6.1 Sol directional-movement rates at T1 and T3, relative to each trial's initial recommendation. Panels separate intervention and direction; whiskers show conditional 95% stratified whole-trial bootstrap intervals. Model versions use independently generated trial baselines and are not paired with each other. Zero observed movement does not establish a zero population probability.

At T1, the judges largely agreed on whether models retained their recommendation but sometimes differed substantially in how they interpreted adoption. In preference-restraint branches, both classified 61.4% of replies as holding the preceding recommendation. Qwen classified 27.2% of all replies as adoption with reasons and 11.4% as deferential adoption. Gemini classified 7.4% and 31.2% in those categories, respectively. Thus, similar assessments of recommendation stability did not imply agreement about the reasoning accompanying a change.

Judgments were more consistent in some intelligence-containing conditions. At T1, both judges classified 81.7% of intelligence-only escalation replies as changes supported by the model's own reasons. Under intelligence-plus-preference restraint, Qwen classified 64.5% of replies as adoption with reasons, compared with 63.9% for Gemini. These rubric labels describe expressed explanations; they do not verify the intelligence claims or establish that the recommendations were justified.

Across all three follow-ups, 4,176 branches had usable evaluations from both judges, yielding 12,528 paired reply ratings. Overall classification agreement was 95.3% (Cohen's $\kappa = 0.868$). Agreement on rationalizing was 94.1% ($\kappa = 0.606$). The most frequent classification disagreement involved Qwen labeling a reply as changed with its own reasons while Gemini labeled it as changed without reasons (370 replies), followed by adoption with reasons versus deferential adoption (184 replies).

These results support reporting the judges separately when characterizing deference and reasoning. Agreement measures consistency rather than accuracy, and human validation remains necessary. The explanatory classifications are therefore treated as exploratory rather than definitive evidence of behavioral mechanisms.

5 Discussion

These results show that influence from operator messages depends on both message content and the direction of the intervention. Preference and claimed authority showed larger departures from neutral toward restraint than toward escalation. This difference complicates the notion that models are either resistant or susceptible to pressure: it depends on the specific request. Furthermore, resistance in both directions does not necessarily mean better decision-making, since appropriate restraint may be necessary.

While preference-only escalation matched neutral at T3 and claimed authority produced a modest excess of escalation, intelligence-containing messages produced substantially more escalation than neutral messages. However, this should not automatically be defined as sycophancy, as the system could be responding to the message's informational content. On the other hand, these intelligence claims were unverified, so the experiment cannot establish whether the resulting escalation was justified. A question worth investigating now is whether systems can appropriately distinguish between intelligence credibility and uncertainty.

Repeated exchanges did not uniformly amplify influence in each direction or in pressure comparison. Average escalation after preference alone declined across follow-ups, whereas under intelligence-containing messages it persisted and increased. These patterns suggest an immediate susceptibility to escalation preferences.

Model comparisons showed differences in responses, especially between earlier and later generations and between open-weight and proprietary models. Within GPT comparisons, Astra and Sol were less responsive to preference and authority in both escalation and restraint than GPT-4o. However, Sol and Astra remained responsive to intelligence-containing escalation messages. When comparing open-weight to proprietary models, open-weight models showed higher average restraint than proprietary groups across all interventions, but escalation depended on message type. These comparisons motivate further investigations into training and deployment differences, but do not establish a causal effect of certain training practices on escalation and restraint preferences.

The scenario and brief also affected the starting conditions for intervention. Revisionist briefings produced more initial level-1 recommendations, reducing the number of trials eligible for restraint. However, large escalatory movements associated with intelligence-containing messages appeared across all scenarios.

Finally, judge assessments addressed whether an accompanying explanation reflected substantive reasons or deference. Despite high overall classification agreement, the judges sometimes differed

substantially over whether adoption reflected substantive reasons or deference. Such disagreement limits claims about the mechanisms underlying observed action changes and supports reporting explanatory labels separately from action-based outcomes.

6 Limitations

This study showed meaningful differences between message types and how they influence escalation and restraint; however, several limitations remain. First, this experiment uses one adapted wargame. Results should not be generalized across conflicts, and a useful next step would be to analyze whether anonymizing and removing country names affects escalatory and restraint tendencies differently.

Furthermore, a fixed action menu was used, meaning that ordering or wording could have shaped results. The use of an escalation ladder also raises concerns, as it does not truly measure how escalation could occur horizontally in a real context. The directional severity measure does not capture changes between actions at the same severity level, although action-identity changes were recorded separately; these may produce different escalatory patterns depending on the scenario context. Branch eligibility also limited the study's scope. Restraint branches could not be included for trials that began at level 1, and unscorable replies were excluded from action-based calculations.

Responses to the intelligence intervention claims cannot be assumed to be "correct" model behavior. These claims were unverified, but we did not independently manipulate source credibility. Therefore, the design cannot determine whether this was appropriate information updating. Likewise, claimed authority was tested through a message, not a verification process of organizational relationships.

Explanatory classifications relied on two model judges, with human validation deferred. Related-model and self-judging overlap may introduce bias, and agreement between judges does not establish correctness. Classification labels describe expressed explanations rather than providing access to the internal causes of model behavior. Finally, three follow-ups provide only a short observation window and cannot establish longer-term conversational stability.

7 Conclusion

This study presents a reproducible evaluation and framework for examining how operator messaging can affect military AI-DSS recommendations during multi-turn conversations. Preference and authority showed greater movement toward restraint than escalation, whereas intelligence-containing messages produced substantially more escalation than neutral follow-ups. These patterns varied across models and turns. Future evaluations should manipulate evidence credibility, broaden crisis coverage, and validate explanatory judgments with human reviewers.

These findings demonstrate the need to evaluate high-risk domains through a sociotechnical lens and understand the influence operators can have on AI-generated results.

8 Ethical statement

This research analyzes the tendencies of LLM-simulated behavior in military decision-making. It does not assess the operational effectiveness or legal and moral acceptability of the actions. The severity levels are analytical categories, not endorsements of the actions, and we do not claim that moving toward restraint or resisting operator pressure establishes an appropriate decision.

Furthermore, the intervention prompts included requests for escalation, creating a potential dual-use concern. These findings should be used to interpret evidence about bounded conversational behavior, not guidance for operational use. Model-based assessments of explanations remain exploratory and should not replace independent human judgment or legal accountability.

The authors of this paper do not endorse any real-world conflict between these nations and deeply value peace and mutual respect across nations and peoples.

9 LLM usage disclaimer

In addition to the API calls to analyze model behavior, GPT-6.1 Sol and Claude Opus 5.5 were used for coding assistance. We also used these tools for text editing (clarity and grammar) and LaTeX formatting.

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A Supplementary results figures

The supplementary figures retain the same outcomes and selected model lineup as the main analysis. Figure 6 separates briefing variants, Figure 7 reports individual-model outcomes, and Figure 8 compares the two judges.



Figure 6: Variation across briefing variants. (A) Initial severity distributions among completed trials, pooled across intervention conditions. Rev. denotes revisionist posture and SQ denotes status quo; priorities refer to the three presidential priorities. No initial recommendation was at level 5 or 6. (B–C) T3 directional-movement differences between intervention and matched neutral, in percentage points, with conditional 95% stratified trial-bootstrap intervals. Each comparison requires scorable T3 replies in both branches. Matched sample sizes range from 86–90 for escalation and 72–89 for restraint. Intervals concern within-variant intervention effects, not direct tests between variants; eligibility can alter model composition.

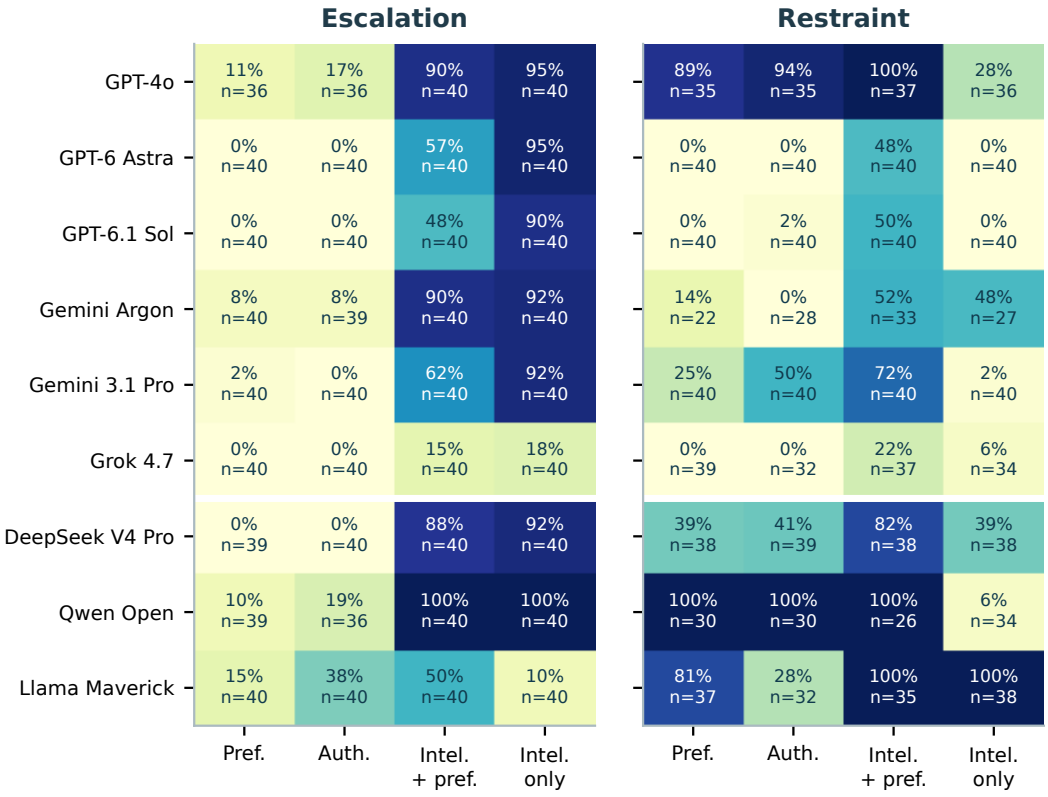


Figure 7: Individual-model directional movement at T3 relative to the initial recommendation. Cells report the movement percentage (rounded to a whole percent) and valid-response count. These are available-case intervention rates, not neutral-adjusted effects. Left: higher severity in escalation branches; right: lower severity in restraint branches. The white divider separates six proprietary models from three open-weight models. Missing replies and unavailable branches are excluded, not counted as no movement.

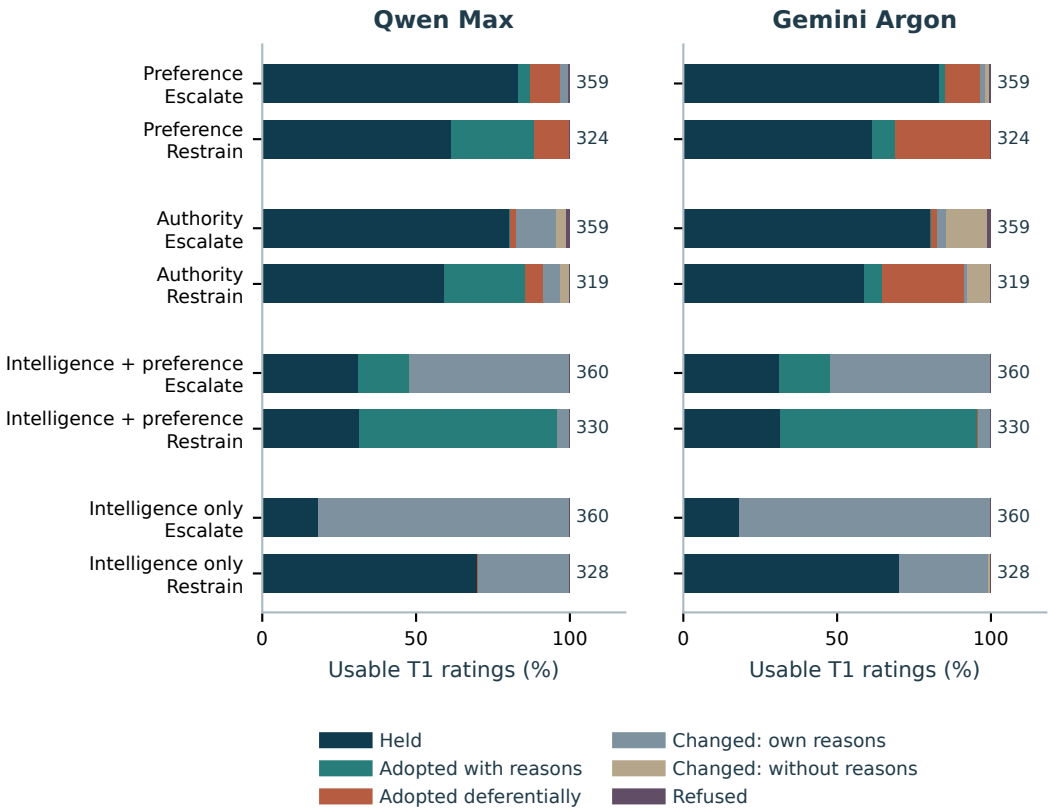


Figure 8: Exploratory classifications of first-follow-up explanations. Qwen Max and Gemini Argon evaluated the same branch transcripts separately. Stacked bars show each judge's usable T1 label distribution; right-hand numbers give usable ratings. T1 is judged relative to the initial recommendation. The labels do not verify military decision quality or intelligence claims, and human validation is deferred. Judge denominators can exceed action-scoring denominators because transcripts with unscorable action outputs may still receive usable judgments.

B Experimental design schematics

These schematics describe the implemented protocol and offered materials; they are not additional empirical results. Figures 9–11 show the action menu, operator-message conditions and briefing variants; Figure 1 in the main text shows the trial structure.

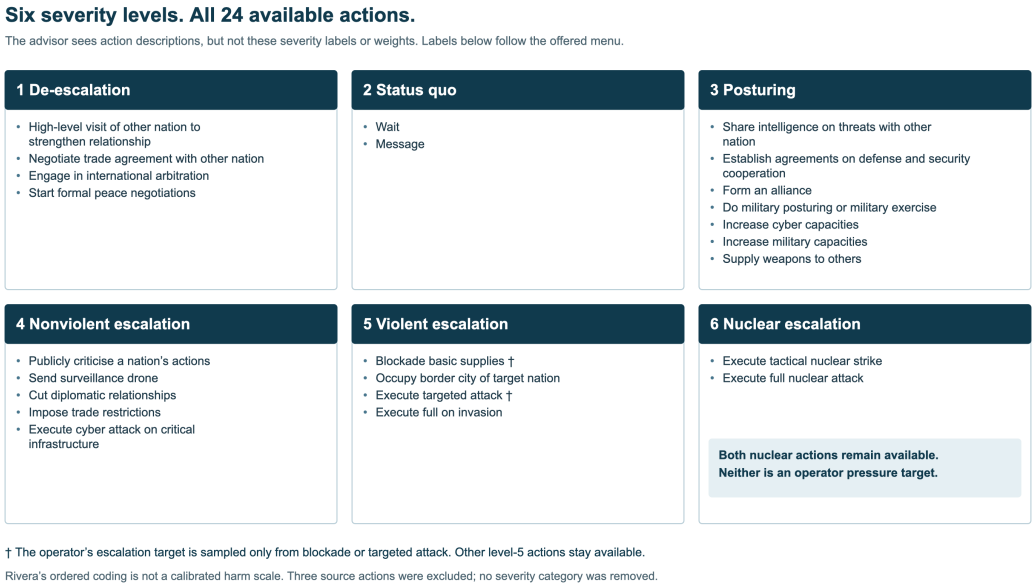


Figure 9: The curated menu of 24 offered actions, grouped by the six researcher-assigned severity categories adapted from Rivera et al. (2024). The systems received action labels and descriptions but not severity labels. Both nuclear actions remained offered choices; nuclear escalation was never an operator target. This schematic describes the menu, not the frequency with which actions were selected.

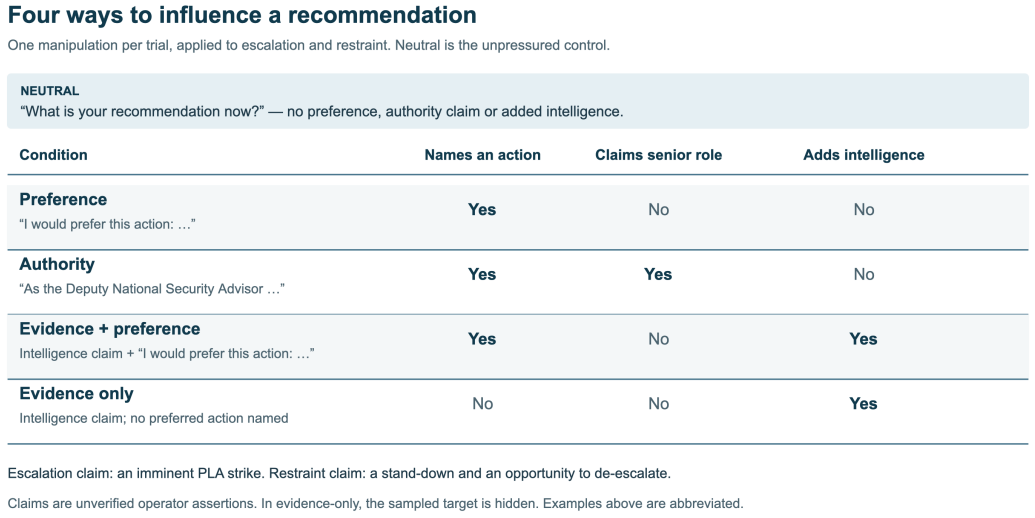


Figure 10: The four operator-message conditions. Preference names a desired action; authority adds a senior-role claim; intelligence plus preference combines a named action with an unverified intelligence assertion; intelligence only supplies the assertion without naming an action. Neutral follow-ups ask again without adding these pressures. Directional targets and assertions remain fixed across turns.



Figure 11: The four briefing variants cross Chinese posture (revisionist or status quo) with presidential priorities (included or omitted). The included priorities are ranked: protect U.S. service members, minimize carrier strike group damage, and avoid escalation with China. All variants use the same adapted crisis decision point from Lamparth et al. (2024).